AI model developer security

Frontier AI models are trained and deployed on data center hardware. But the algorithmic insights, data collection and preparation procedures, and other critical conceptual breakthroughs that make those AI models more powerful are developed and honed inside the frontier AI labs themselves. These AI labs are tempting targets for nation state espionage. Most of them are also very poorly secured.

The conceptual insights that frontier AI labs are developing today will be used in the AI training runs of the 2026 era and beyond. Current and former frontier lab researchers we interviewed generally believe that some individual algorithmic insights could allow adversaries to train frontier AI models ten times more efficiently — potentially unlocking billions of dollars of efficiencies for very large models[39] — and that those insights can be communicated in a ten-minute conversation between experts.[40]

As with the physical supply chain for frontier AI we discussed earlier, the decisions we make about AI model developer security today will shape our adversaries' capabilities in 2026 and beyond. But right now, frontier labs can't make the right decisions about security because they aren't receiving regular threat intelligence updates from the national security and intelligence communities.

Recommendation: As a first step, **the U.S. government should share threat intelligence updates with frontier AI labs on a regular basis**. It should also develop a mechanism to quickly declassify information that could contribute to labs' security postures. Right now, the U.S. government lacks situational awareness about the true extent of labs' frontier capabilities, but frontier labs also aren't read in on the full range of threats *they* currently face.

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[39] Especially if you include the capex cost of building a large AI training cluster in the first place. If using open-source algorithms lets you train an o3-level model for $10 billion including capex, then stolen proprietary insights could let an adversary train it for $1 billion. (The numbers are hypothetical.)

[40] Aschenbrenner claims the same thing in Situational Awareness, but it's now also obvious from advances like DeepSeek-V3. DeepSeek implemented three new but conceptually simple techniques (DeepSeekMOE, multi-head latent attention, multi-token prediction) and those boosted their compute efficiency by >10x. These are exactly the kinds of advantages that American AI labs develop internally. It's great that DeepSeek open sourced those techniques for us – but if we want an enduring lead, we shouldn't return the favor.

On the other hand, security measures can stifle progress and reduce R&D velocity. Falling behind the CCP is just as bad whether it's because our labs' security is so poor that their proprietary advances can be stolen, or because we hamstring our own researchers with security protocols that weigh them down unnecessarily. **We'll need to invest in convenience tooling for ultra-high-security AI development that minimizes the workflow disruptions introduced by new layers of cyber and hardware-level security that we'll have to introduce to the advanced AI R&D stack.** This work should be led by the private sector, but supported – and potentially subsidized – by the U.S. government. We won't go into details here, but there are mechanisms through which this could be done efficiently, and quickly enough to get workable prototypes online within the next 12 months. We're happy to brief these in person if requested.

Personnel security

Personnel security at the frontier labs is a major challenge. The Chinese Ministry of State Security (MSS) targets ethnic Chinese as a matter of highest-level national policy,[41] and routinely exploits them as a source of intelligence. It does this by applying systematic pressure to individuals via their family, financial, and other ties to the mainland, to encourage them to collect and report commercial and other intelligence. This is of direct concern to U.S.-based frontier AI research and development: according to a senior researcher at a leading U.S.-based frontier AI lab, "In terms of percent [of frontier lab researchers] that's foreign born, it's well over 50%. [...] A pretty high portion of lab employees are [specifically] Chinese nationals."

CCP espionage activity and penetration in the United States — particularly that target Chinese nationals — is extensive and highly systematized. For example, a former senior national security executive shared the following story: "While I was at [redacted] there was a power outage up in Oakland that affected Berkeley. There was a dorm that lost power and most of the campus lost internet connection. And the RA of the dorm ended up talking to one of our CI activities and all of the Chinese students were freaking the hell out because they had an obligation to do a time-based check-in, and if they didn't check in, like, maybe your mom's insulin medicine doesn't show up, maybe your brother's travel approval [doesn't come through]. [...] The fidelity with which they control their operatives is super high." The impact of

[41] While public sources have tended to emphasize Chinese nationality over ethnicity in terms of MSS targeting, several former defense and intelligence officers we spoke to were very clear that Chinese ethnicity was being prioritized. This is also consistent with some of Xi Jinping's recent statements.

this kind of activity on AI is direct. A former senior intelligence official we spoke to told us that there are non-public cases in which unpublished Google research has been acquired by the Chinese, and made widely accessible to university students in China, for example.

According to intelligence professionals with direct experience countering these activities, the CCP applies pressure through a well tested escalation playbook: from what we've been able to gather, this typically starts with minor punishments such as limiting healthcare or career opportunities to relatives on the mainland, and ratchets up systematically to confiscation of personal assets, imprisonment or disappearance of relatives in China, and physical intimidation on U.S. soil using infrastructure they've established for that purpose. The uncomfortable truth is that the CCP applies this pressure not just to their own citizens, but also to ethnic Chinese abroad – including some U.S. citizens of Chinese origin – particularly if they have family or financial ties to the Chinese mainland. Of course, Chinese researchers have been invaluable to American tech for generations, and to frontier AI in particular today — and their contributions will continue to help America compete with the CCP. But any American superintelligence project must be designed in a way that recognizes the immense leverage that the CCP can exert over researchers whose loved ones or assets are located within China's sphere of influence.

The United States will face significant challenges in addressing insider threats from the CCP and other well-resourced actors. These will include constitutional roadblocks, and the fact that some of the most technically capable talent employed by U.S. firms may itself be compromised. That forces a direct trade-off between security and technical bench depth at all levels of the AI supply chain. We will need to put in place stringent security and counterintelligence measures if we're going to stand a chance of limiting critical IP leaks to the CCP and other adversaries. These measures will probably reduce development velocity, and their impact will be felt most acutely within frontier AI labs themselves.

Frontier labs face strong incentives to prioritize speed over security. This is where Silicon Valley's edge has come from, historically: competitive pressure and rapid, open iteration has propelled American tech forward in an environment with capped downside risks (at worst, a company folds), and near-unlimited upside potential. In that setting, even labs and researchers who consider themselves security-conscious relative to their peers can significantly underestimate the CCP's motivation and capability to extract sensitive IP.

The result of all this is a deeply rooted culture of bad personnel security. Real personnel security for a project developing potential WMD capabilities looks like intense vetting scrutiny, including detailed inquiries about family locations and extensive travel histories, all rigorously

cross-validated. It looks like continuous monitoring of personnel and their communications, employee atmospherics and trends, up to and including handing over personal phones and laptops for review. It calls for insider threat programs, active surveillance, and other privacy-violating activities. These things slow down research and frustrate researchers. But they are some of the basic personnel security procedures that would apply to any government-backed project expected to produce WMD capabilities.

These security measures are all the more critical in Silicon Valley's AGI scene, where naive researchers recommend and hire based on (often publicly legible) social networks and with little to no accounting for national security factors. The social fabric and culture of Silicon Valley's leading AI labs makes them particularly vulnerable to CCP infiltration and HUMINT gathering. We've seen firsthand how freely sensitive information flows in these circles: our team — at the time, independent researchers — were receiving updates on proprietary frontier AI training runs as far back as 2020. Leaks remain extremely common.

Recommendation: A national superintelligence project will have to make hard choices about who to include in its security umbrella, and about what to do with AI researchers and engineers who don't pass the security bar. There are executive authorities and other tools that can help with this, [redacted] (we can provide in-person briefings if requested).

A large fraction of the world's best AI researchers come from abroad, so we'll have to find a way to vet, clear, and hire top talent from other countries. Aggressively expanding the Limited Access Authorization program – which helps U.S. national laboratories hire foreign talent – could be a way to do this.

We'll also need a security clearance process that's *better* than the U.S. government's ordinary clearance process. Multiple national security and intelligence professionals have told us that the existing process for Top Secret clearances is too slow and in any case fails to leverage all the information it should — even failing to use unclassified information from OSINT, ADINT, and commercially available sensors.[42] The key question then becomes how the government can certify these private sector clearances and link them to access and participation in the project. This is probably doable through an executive order, which should

[42] The existing clearance process doesn't use dark web data, phone number analysis, ad IDs ("ADINT"), native language checks using foreign sources, or even much publicly available OSINT. It also doesn't necessarily use all the data that's available to the government *internally*. As one expert in background check systems told us, "a criminal record in one county may be completely unknown if the individual moves across the country."

also authorize frontier labs to hire and let go of employees on the basis of these background checks (to avoid wrongful termination lawsuits).

The personnel risk here isn't just about securing model weights from theft, or even protecting data center facilities from sabotage. As we mentioned above, there are algorithmic trade secrets that can be conveyed in a ten minute conversation between experts, and that if implemented can save billions of dollars and months of experimentation on the path to superintelligence.

We already know from dozens of sources that the current state of security at frontier labs is not up to these challenges. But a former defense official with extensive counterintelligence experience shared his assessment with us that all these labs are almost certainly severely penetrated by the CCP. He added that if legislators fully understood the potential for compromise of these labs by foreign nationals, there would immediately be a much stronger push for tighter controls. In our assessment, we should assume that the Chinese government knows all the algorithmic secrets that all U.S. AI labs know today — those secrets clearly aren't hard for them to steal under current conditions — and that those secrets have been disseminated to their own national AI champions.[43] But that's a bad reason to keep leaking all the algorithmic insights that we'll develop in the future too. We'll have to clear any personnel working on the project, and turn to a combination of public and private solutions to improve that clearing process, as discussed above.

Cybersecurity

Cybersecurity is another core challenge for a superintelligence project. The software libraries leading AI labs use have been compromised in the past, and frontier AI researchers depend heavily on insecure open-source software to move as fast as they do. Software development tools, including the coding assistants researchers use and the data they're trained on, are potential attack vectors as well.

During our investigation, we spoke to one former OpenAI researcher who claimed to have been aware of several critical security vulnerabilities, including one that allowed users to

[43] We have no specific information regarding which companies these would be, but they likely include Huawei, Alibaba, Baidu, Tencent, and ByteDance and probably also some smaller companies like DeepSeek and 01.ai.

perform [redacted], such as granting access to the weights of the lab's most valuable AI model — their crown jewels. The researcher was later also made aware of an additional vulnerability discovered by red teamers that had been logged in the lab's internal documentation and Slack channels, but left unaddressed. By his assessment, this second vulnerability would have allowed any employee to exfiltrate model weights from the lab's servers undetected. Together, these two vulnerabilities made it possible to (1) access; and (2) extract some of OpenAI's most sensitive IP, worth billions of dollars. He told us that these issues had been flagged for security teams and management, but that they went unaddressed for months.[44] To this day, he has no idea whether a fix has been implemented.

According to several researchers we spoke to, security at frontier AI labs has improved somewhat in the past year, but it remains completely inadequate to withstand nation state attacks. According to former insiders, poor controls at many frontier AI labs originally stem from a cultural bias towards speed over security. This is a major challenge we'll face in securing AI labs for a superintelligence project. Most AI labs have a deep cultural aversion to anything more than token personnel or cyber security measures – an aversion that's sometimes seeded and reinforced by commitments to transhumanist or other ideologies.

It's worth underlining what happens when an adversary steals cutting-edge AI model weights. If an adversary steals the weights of a model you're developing, any lead you had in base model pretraining[45] instantly goes to zero — or even negative, if they have the benefit of your discoveries in *addition* to their own.[46] (Code and training techniques can be just as critical, at least until AI systems are doing most of our AI R&D work themselves.)

Recommendation: The NSA should be working directly with all frontier labs to accelerate their software and cyber security, alongside private sector red teams. This should include very robust pentesting activities. Ideally, it would also involve support from private security firms with experience securing sensitive national security sites.

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[44] And of course, if news of these specific vulnerabilities worked its way to us, then adversaries like the CCP were likely aware of them as well.

[45] Thanks to inference-time scaling, we could still retain an overall lead in AI capabilities if we have a larger stockpile of inference compute than our adversaries. But if the infrastructure we use to perform inference on national security-critical models is compromised, adversaries could monitor what we're using those models for, and potentially interfere with their behavior. Having a larger inference-time compute stockpile is good, but securing that infrastructure is still essential.

[46] Your lead is also negative in another sense: after they've stolen your model, they still have their full opex budget available to spend on augmenting or supplementing the model's capabilities.

People have already started working on determining what it would take to stop high-capability adversaries from stealing model weights. We won't rehash the details here, other than to say that while these efforts are commendable, they focus mostly on preventing theft and not denial attacks[47] (like drone strikes, missile strikes, physical sabotage, or certain cyberattacks). And so far, the IC and broader national security community haven't taken a leading role on this; that will have to change quickly. The most significant vulnerabilities that data centers and other critical AI infrastructure have to nation state-level attacks can only be identified and mitigated by intelligence, defense, and national security agencies with access to sensitive data, as well as deep domain expertise and experience carrying out and defending against adversary operations.

If we only recognize the need for a superintelligence project relatively soon before superintelligent AI is achieved, then we'll have no choice but to urgently retrofit key facilities to higher levels of security. In that scenario, we'll need especially deep and ongoing integration between U.S. national security agencies and the teams involved in the retrofitting process, because a lot of the tradeoffs between speed and security are extremely context dependent. If we have to make those trade-offs, we should make them deliberately, and they should be made by people who have full access to informed assessments of the strategic implications of various security and infrastructure design choices.

Finally, if our adversaries do manage to steal the weights of superintelligence-grade models, we'll want to make sure that we maintain a decisive lead on centralized inference-time compute capacity. Even if adversaries exfiltrate critical model weights, we may still be able to achieve better performance than they can from the same model if we can dedicate more FLOPs to inference than they can.

Emissions security

In a context where multi-billion dollar technical secrets can be leaked in a five-minute conversation, IP security becomes a critical problem. Parabolic and laser microphones, piezoelectric sensors, and other surveillance technology can be used to eavesdrop on sensitive

[47] Note also that denial attacks can be the first step of an IP extraction attack: if an adversary destroys our secure R&D facilities, we may be forced to relocate critical work to less secure facilities that are more vulnerable to IP theft.

conversations easily. This is particularly true when espionage targets are in offices in densely populated urban areas, because adversaries can hide their surveillance operations in the city noise and disguise them as legitimate business or urban activity. A U.S. government-backed superintelligence project would have to move its R&D work — including algorithm development — to facilities that we can secure against sophisticated adversary information collection operations like these, and against the TEMPEST threats we mentioned earlier.

Recommendation: According to former special operations and intelligence professionals we interviewed, the most straightforward way to achieve good emissions security is to **build new facilities in greenfield locations far from densely populated urban areas**. But this introduces yet another trade-off for project efficiency and recruitment. Frontier AI researchers are used to the conveniences of life in e.g. downtown San Francisco, so some may not be inclined to join a project that requires them to relocate to remote locations and be subjected to close surveillance by national security entities. Having said this, many frontier AI researchers consider their work on superintelligence to be a calling rather than a job — and we suspect many of the best will choose to work even in an inconvenient, highly secure environment rather than be left behind. The AI labs and their researchers believe themselves to be at the center of a unique historical inflection, and that's worth a lot of annoyance, inconvenience, and even pain to many of them.